Advanced Computer-Aided VLSI System Design

Final Project:
ResNet Inference Accelerator

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NTU GIEE

Overview

- Residual network (ResNet) is deep learning architecture and it is widely applied to image classification and object detection
- The architecture of ResNet further inspires more powerful deep learning model like BERT or GPT

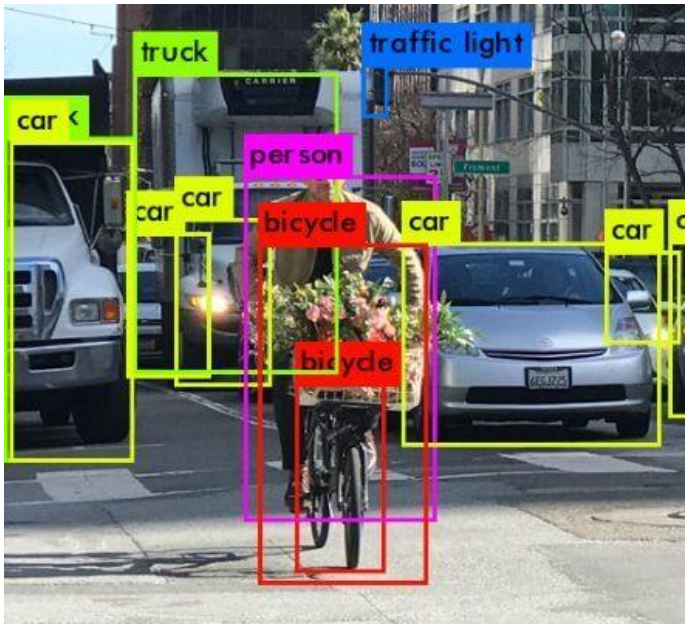


cat



dog

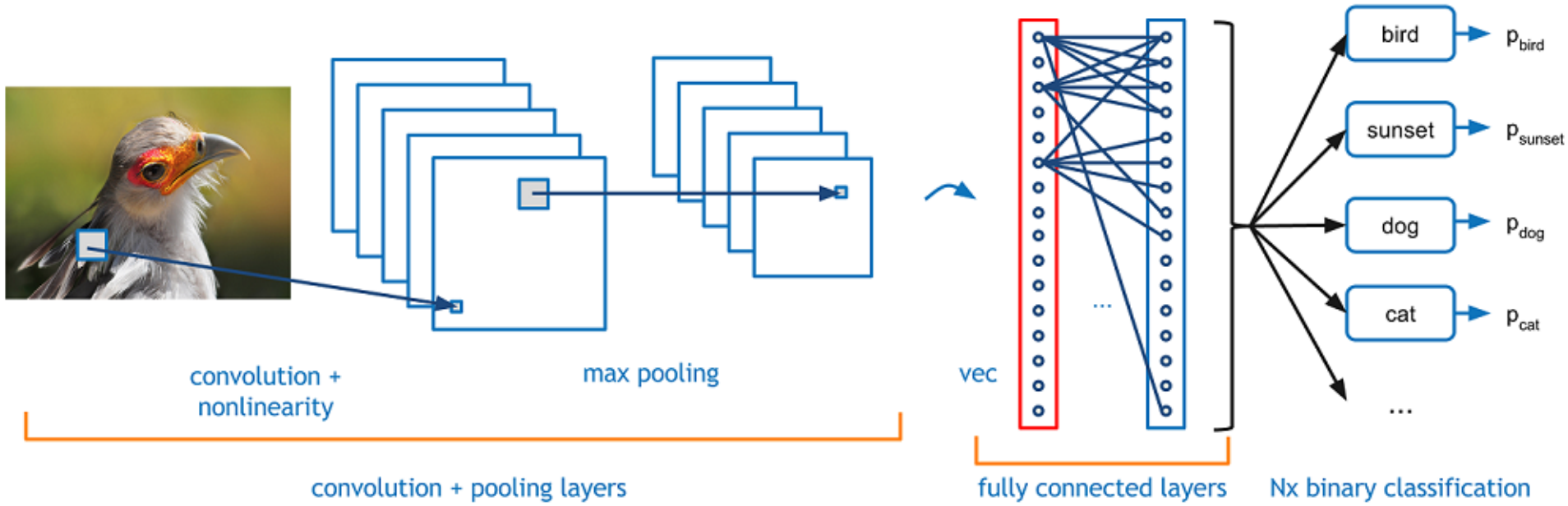
Image Classification



Object Detection

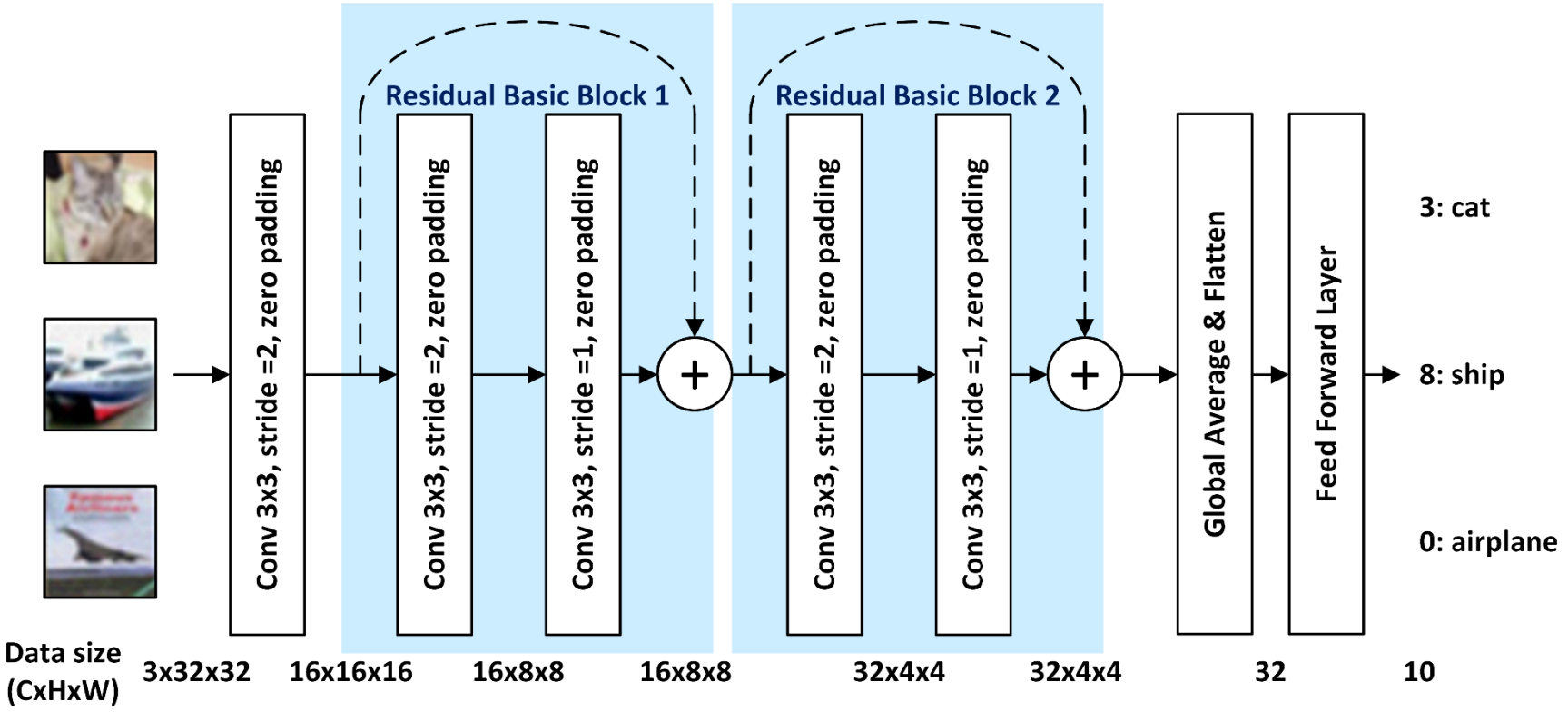
CNN Image Classification

- The CNN model is an expert at image classification
- The category of the image is predicted through the index of maximum value of output of model
- Cafir-10 dataset and ResNet 6 CNN model are used in this project



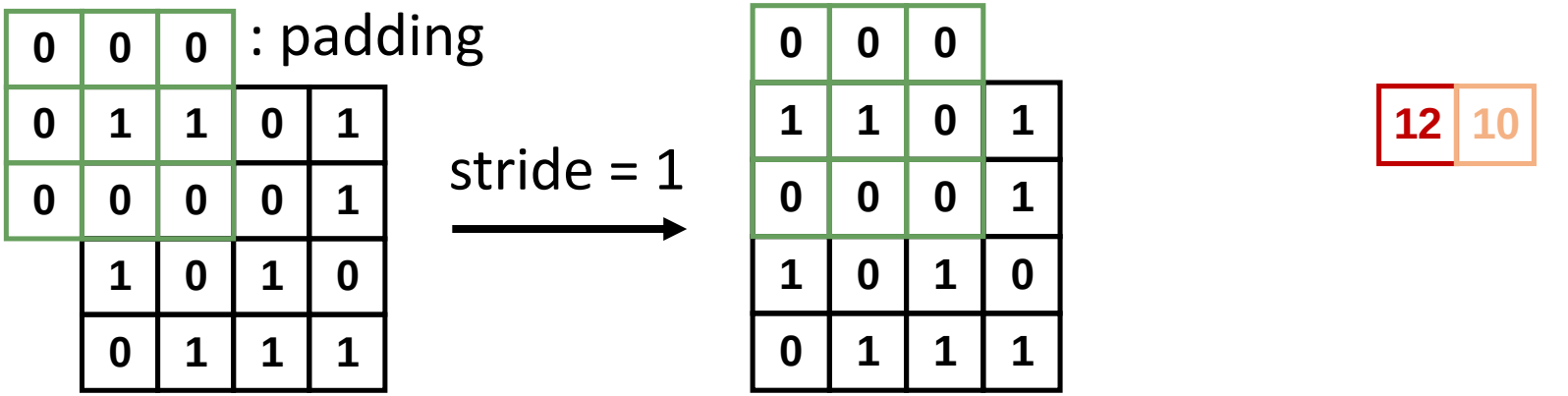
ResNet Architecture

- The image below shows the architecture of the ResNet 6
- The ResNet 6 is constructed by 1 convolution block, 2 basic block, and 1 feed forward layer



ResNet Operation: Convolution

- Convolution is the basic operation of CNN model
- The details of the convolution operation refer to the note of midterm project



= 1x5 + 1x6 + 1 = 12
(multiply and sum, then add bias)

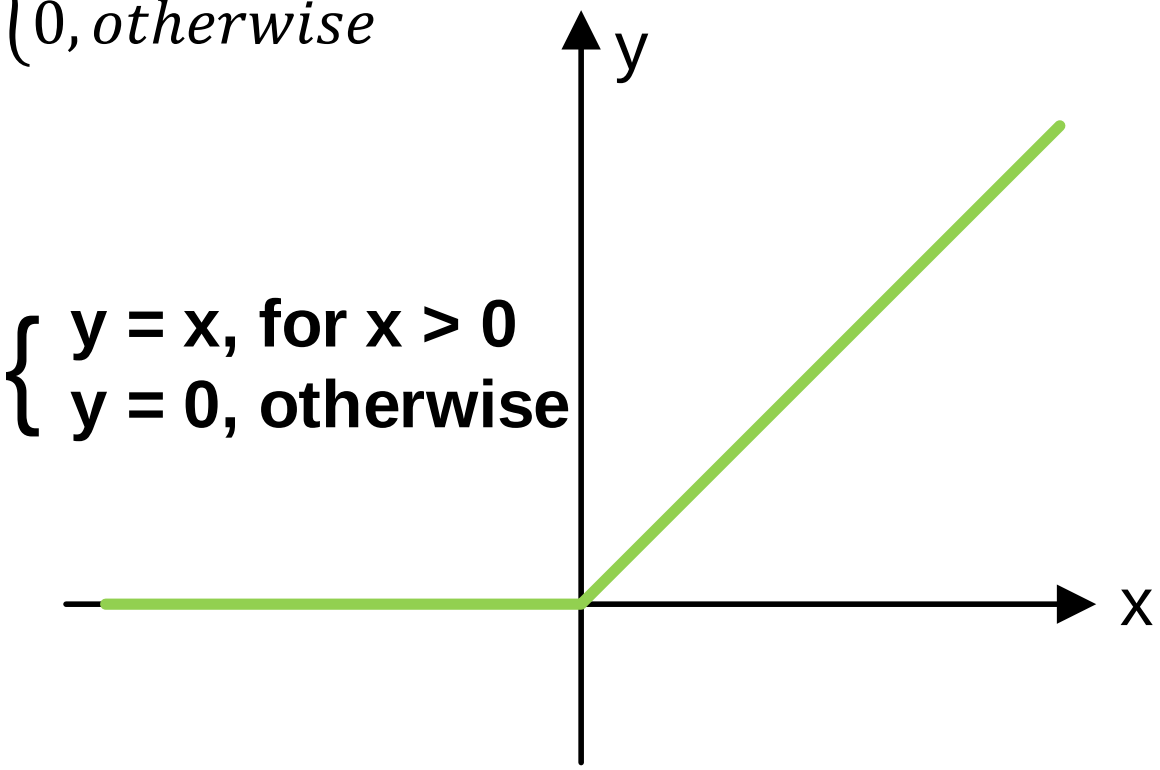
= 1x4 + 1x5 + 1= 10

output

ResNet Operation: ReLU

- ReLU is a type of activation function
- The ReLU formula:

$$- \ y = \begin{cases} x, for\ x > 0 \\ 0, otherwise \end{cases}$$



ResNet Operation: Batch Normalization

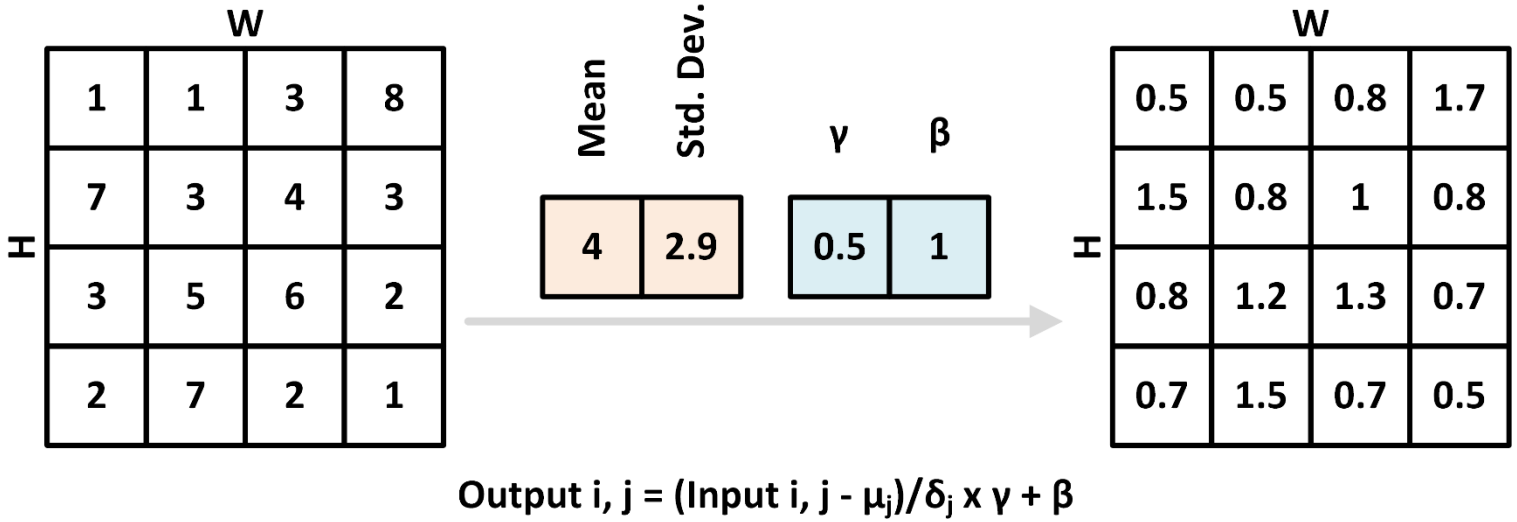
- For 1-dimension batch normalization, the value is modified according to the mean and standard deviation along the batch dimension
  - $out = \frac{in - \mu}{\delta} \times \gamma + \beta, u\ is\ the\ mean\ and\ \delta\ is\ the\ standard\ deviation$
- The mean and the variation are fixed while operating inference

Input 1	Input 2	Input 3	Mean	Std. Dev.	γ	β	Output 1	Output 2	Output 3
1	3	8	4	2.9	0.5	1	0.5	0.8	1.7
3	4	3	3.3	0.5	1	3	2.3	4.4	2.3
5	6	2	4.3	1.7	0.8	1.2	1.5	2.0	0.1
7	2	1	3.3	2.6	0.7	2.5	3.5	2.1	1.9

Output i, j = (Input i, j - μ_j)/δ_j x γ + β

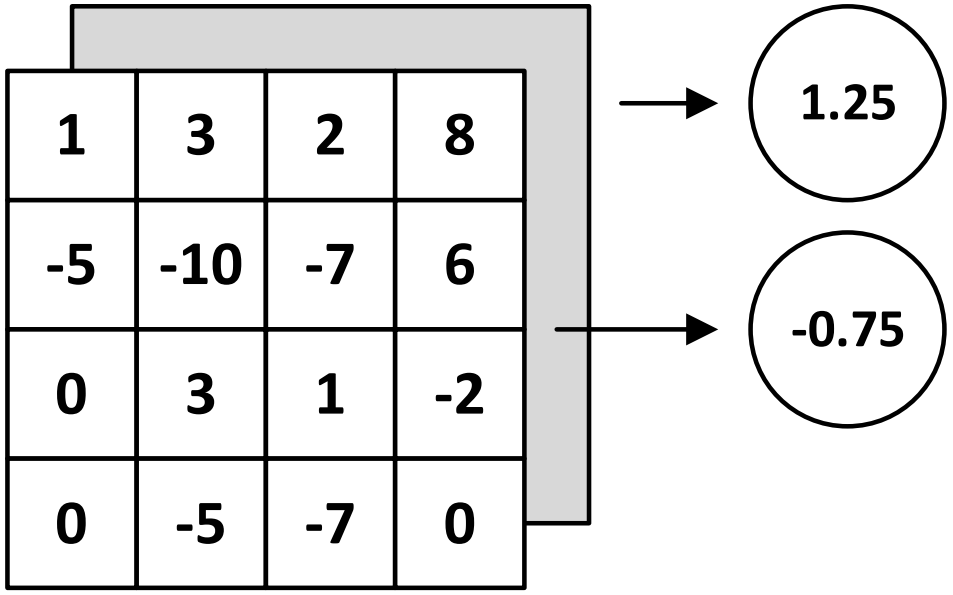
ResNet Operation: Batch Normalization

- For 2-dimension batch normalization, a channel shares the same mean and standard deviation
  - $out = \frac{in - \mu}{\delta} \times \gamma + \beta, u \; is \; the \; mean \; of \; the \; channel$
    $and \; \delta \; is \; the \; standard \; deviation \; of \; the \; channel$



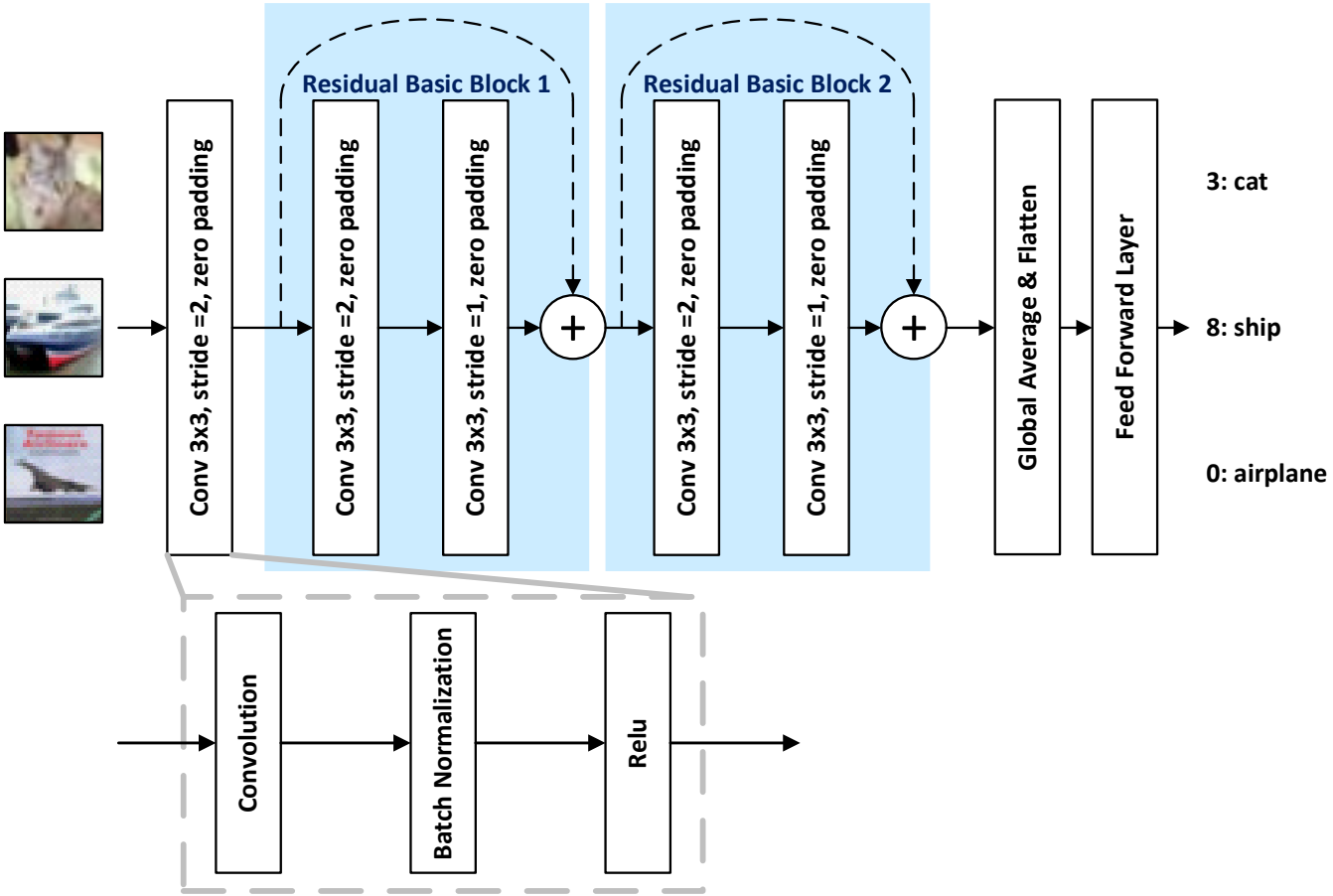
ResNet Operation: Average Pooling

- The average pooling operation computes the mean of the channel
- The 3-dimension input will be collapsed to 1-dimension output after average pooling and flatten operation

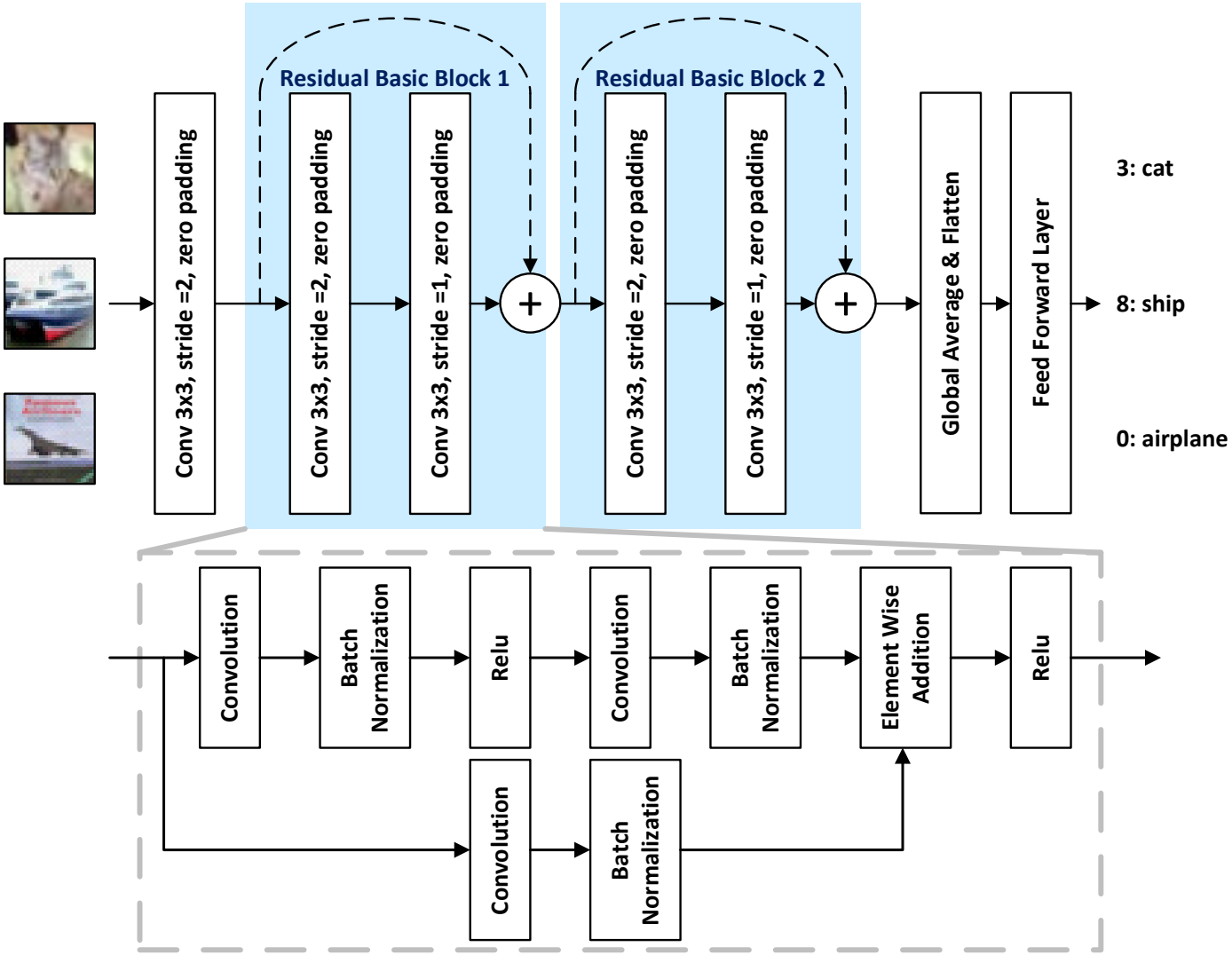


ResNet Operation: Convolution Layer

- A convolution layer contains a convolution operation followed by a batch normalization and a ReLU

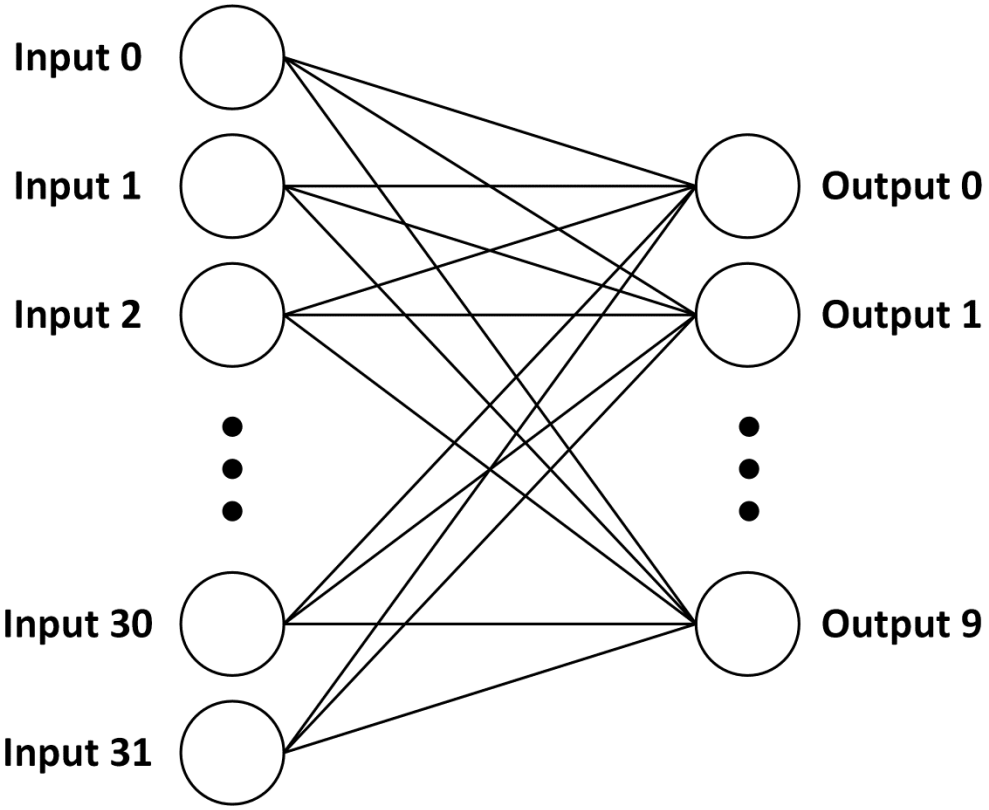


ResNet Operation: Basic Block



ResNet Operation: Feed Forward Layer

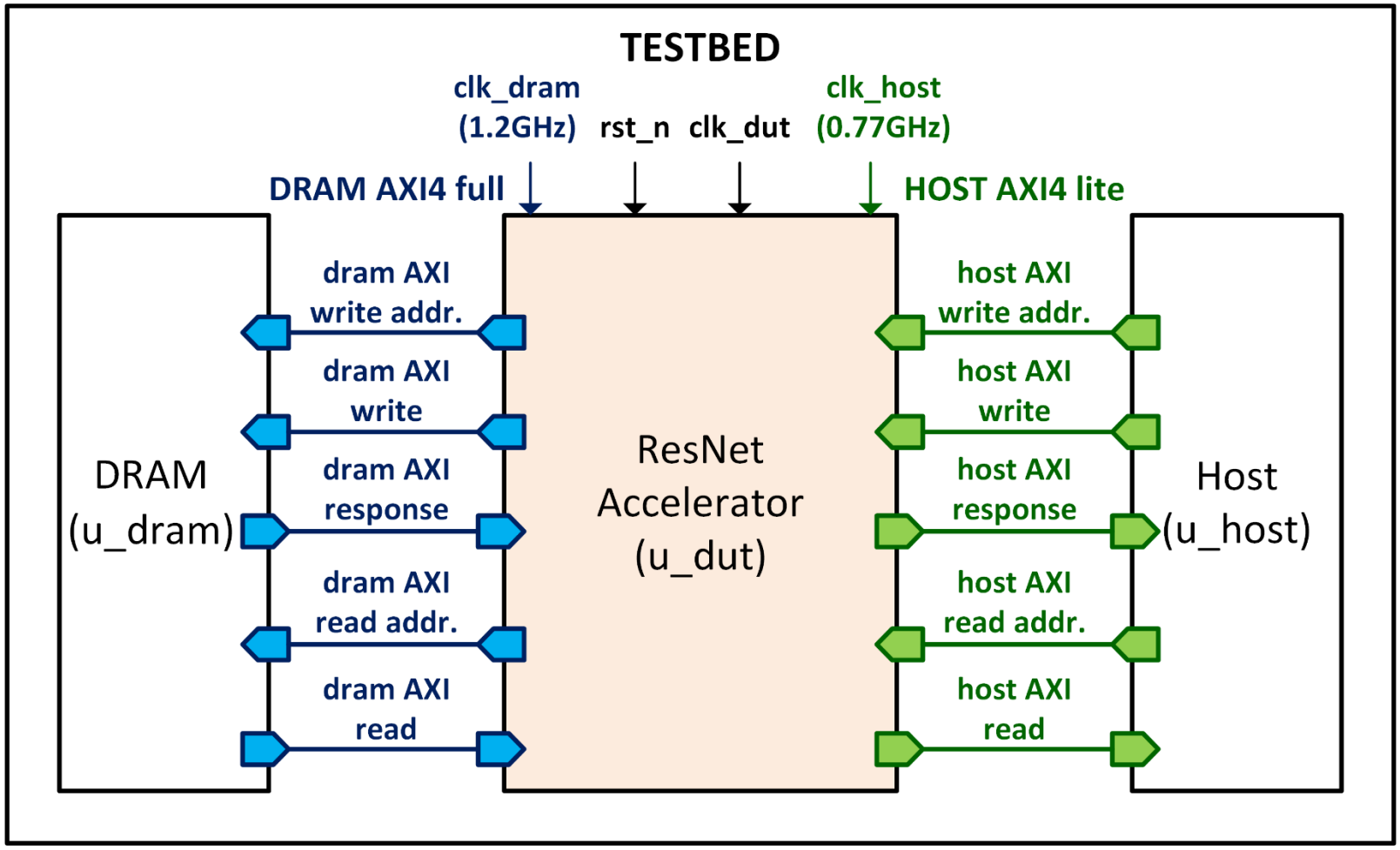
- Feed Forward Layer is a linear vector-matrix multiplication
- Soft-max operation before feed-forward layer is required



Output = Weight x Input + Bias

[10] = [10x32] x [32] + [10]

Block Diagram



Input/Output : Clock and Reset

Signal Name	I/O	Width	Description
`clk_dut`	I	1	The system clock signal for ResNet accelerator You can define the clock frequency (>= 100MHz)
`clk_dram`	I	1	The system clock signal for the signal from/to DRAM The frequency is fixed at **1.2GHz (clock period 0.834 ns)**
`clk_host`	I	1	The system clock signal for the signal from/to host The frequency is fixed at **0.77GHz (clock period 1.3 ns)**
`rst_n`	I	1	**Asynchronous** active low reset signal

Input/Output : Host AXI4 Lite

Signal Name	I/O	Width	Description
host_axi_awaddr	I	8	Address of the first transfer in a transaction.
host_axi_awvalid	I	1	Write address request valid indicator.
host_axi_awready	O	1	Write address request ready indicator.
host_axi_wdata	I	32	Write data.
host_axi_wstrb	I	4	Indicator of which byte lanes of write data contain valid data.
host_axi_wvalid	I	1	Write data valid indicator.
host_axi_wready	O	1	Write data ready indicator
host_axi_bresp	O	2	Response for write channel **(Only 2'b00 allowed).**
host_axi_bvalid	O	1	Write response valid indicator.
host_axi_bready	I	1	Write response ready indicator.

Input/Output : Host AXI4 Lite

Signal Name	I/O	Width	Description
host_axi_araddr	I	8	Address of the first transfer in a transaction.
host_axi_arvalid	I	1	Read address request valid indicator.
host_axi_arready	O	1	Read address request ready indicator.
host_axi_rdata	O	32	Read data.
host_axi_rresp	O	2	Response for read channel **(Only 2'b00 allowed).**
host_axi_rvalid	O	1	Read valid indicator.
host_axi_rready	I	1	Read ready indicator.

Input/Output : DRAM AXI4 Full

Signal Name	I/O	Width	Description
`dram_axi_awid`	O	8	ID of the write transaction.
`dram_axi_awaddr`	O	19	Address of the first transfer in a transaction.
`dram_axi_awlen`	O	8	Total number of transfers in a transaction
`dram_axi_awsize`	O	3	Maximum number of bytes in a transfer within a transaction.
`dram_axi_awburst`	O	2	Mode of the address increment.
`dram_axi_awvalid`	O	1	Write address request valid indicator.
`dram_axi_awready`	I	1	Write address request ready indicator.
`dram_axi_wdata`	O	256	Write data.
`dram_axi_wstrb`	O	32	Indicator of which byte lanes of write data contain valid data.
`dram_axi_wlast`	O	1	Indicator of the last data.

Input/Output : DRAM AXI4 Full

Signal Name	I/O	Width	Description
`dram_axi_wvalid`	O	1	Write data valid indicator.
`dram_axi_wready`	I	1	Write data ready indicator
`dram_axi_bid`	I	8	ID of the write response transaction.
`dram_axi_bresp`	I	2	Response for write channel **(Only 2'b00 allowed)**.
`dram_axi_bvalid`	I	1	Write response valid indicator.
`dram_axi_bready`	O	1	Write response ready indicator.
`dram_axi_arid`	O	8	ID of the read transaction.
`dram_axi_araddr`	O	19	Address of the first transfer in a transaction.
`dram_axi_arlen`	O	8	Total number of transfers in a transaction
`dram_axi_arsize`	O	3	Maximum number of bytes in a transfer within a transaction.

Input/Output : DRAM AXI4 Full

Signal Name	I/O	Width	Description
`dram_axi_arburst`	O	2	Mode of the address increment.
`dram_axi_arvalid`	0	1	Write address request valid indicator.
`dram_axi_arready`	I	1	Write address request ready indicator.
`dram_axi_rid`	I	8	ID of the read transaction.
`dram_axi_rdata`	I	256	Read data.
`dram_axi_rresp`	I	2	Response for read channel **(Only 2'b00 allowed).**
`dram_axi_rlast`	I	1	Indicator of the last data.
`dram_axi_rvalid`	I	1	Read valid indicator.
`dram_axi_rready`	O	1	Read ready indicator.

DRAM Data Structure

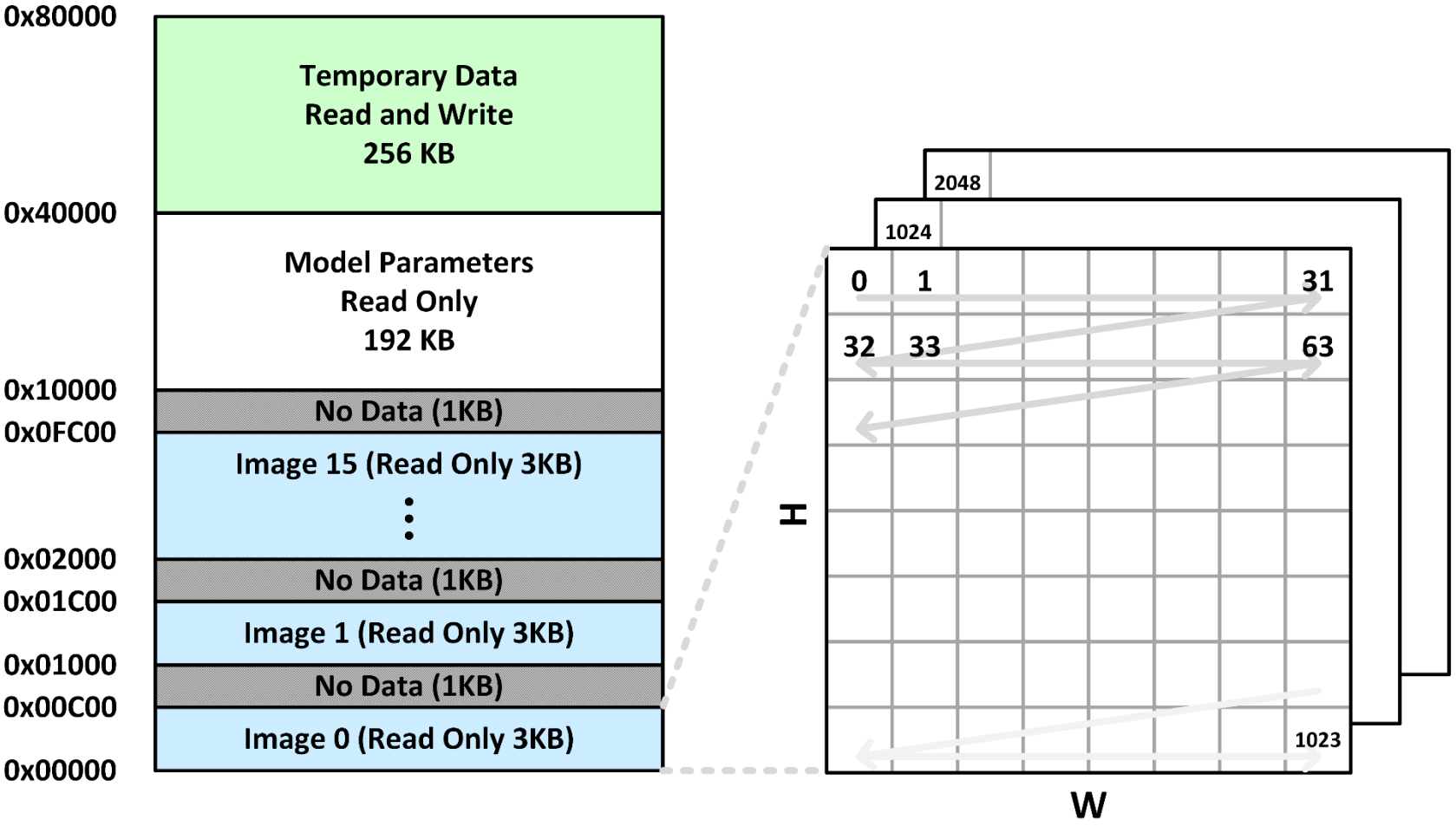
- DRAM contains total 512 KB data
- First **64 KB** data contains **the binary value of images**
  - The images are always placed from the bottom of the DRAM
  - For example, if this batch contains 2 images, the first image is placed at 0x000, and the second image is placed at 0x1000
- The following **192KB** data contains **ResNet Model**
  - You have to construct the data by yourself and it is read only
- The following **256KB** data can put any **temporary data** at the runtime
- There will be some latency of read/write operation
  - Read latency: 12 cycles / Write latency: 16 cycles

DRAM Data Source

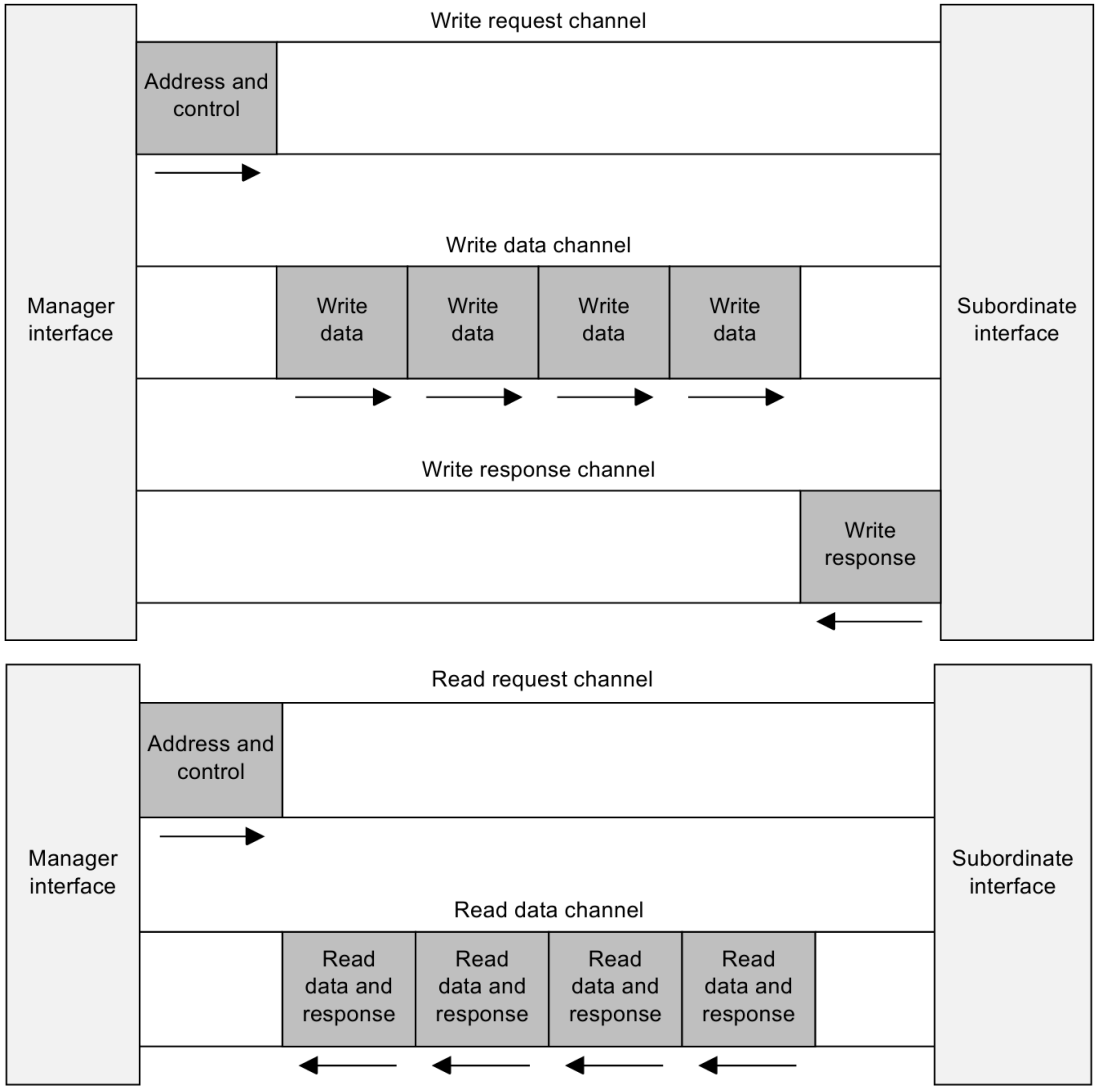
- The binary data of the images is read from 00_TESTBED/PATTERN/IMAGES/image_*.bin at the beginning of each batch
- The model is read from 00_TESTBED/PATTERN/model.dat
  - You should generate it by yourself

```
1   @10000
2   A3 5F 2D C8
3   9B 71 E6 3C
4   48 F2 07 BE
5   61 8E 94 B0
6   2A 56 19 D4
7   77 3B 01 9F
8   EA 6C C1 12
9   88 0E A7 3F
10  45 D2 9C 34
11  A0 5B 8D 7E
12  D1 4C 2E 6A
```

DRAM Data Structure



DRAM Data Transfer

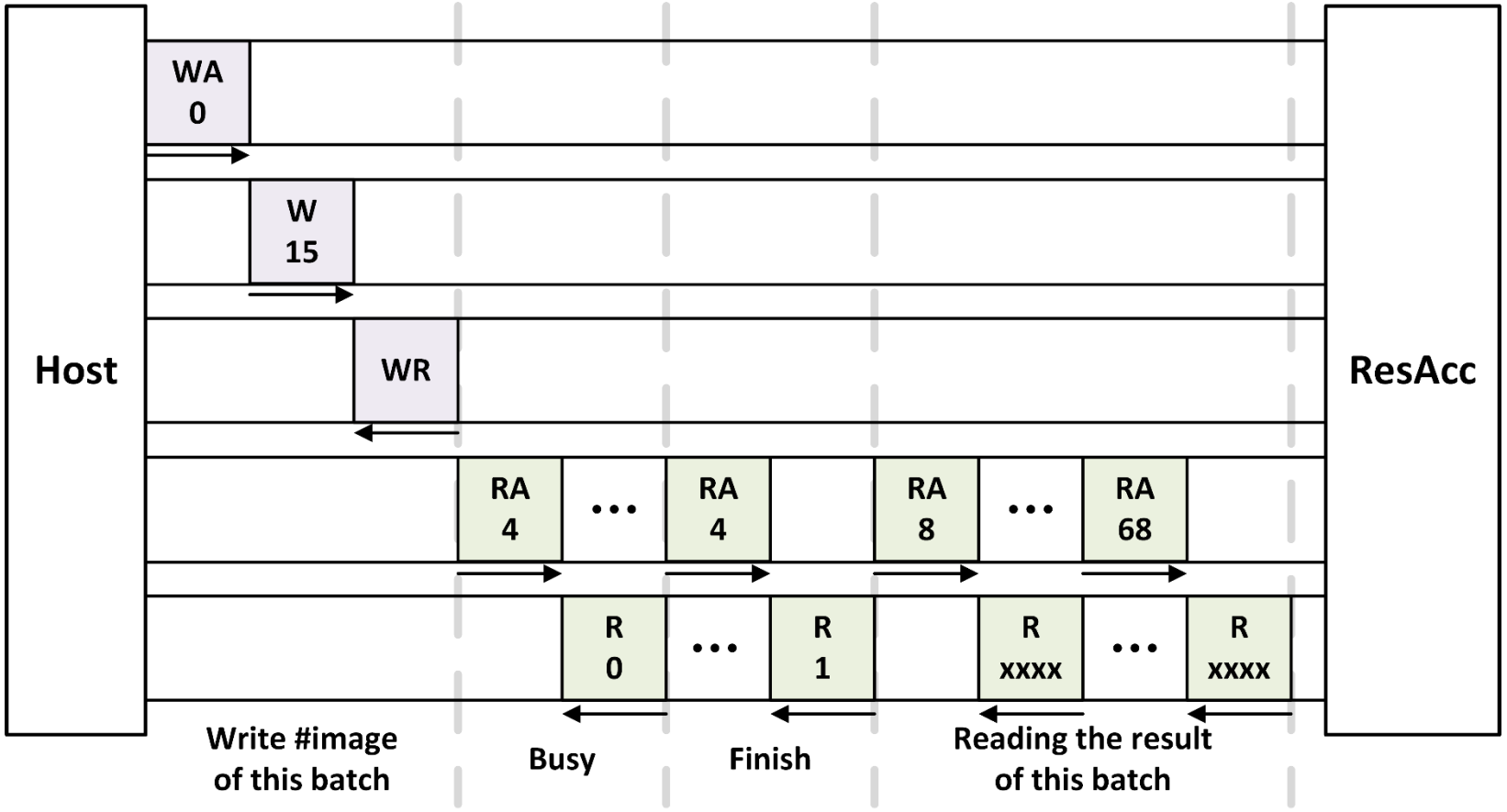


ResAcc Register Structure

- ResNet accelerator (ResAcc) contains 72 Bytes data
- The data represents any information about the batch

Addr.	Access	Data Content (32bit/data)	Data Note
0x44	Read Only	bit: 31-28 category bit: 27-0 confidence	The result of image 15
⋮	⋮	⋮	⋮
0x0C	Read Only	bit: 31-28 category bit: 27-0 confidence	The result of image 1
0x08	Read Only	bit: 31-28 category bit: 27-0 confidence	The result of image 0
0x04	Read Only	bit: 0 finish 1/busy 0	Finish & Busy indicator of this batch
0x00	Write Only	bit: 3-0 number of Image -1	The number of images of this batch (Ex: value 15 for 16 Images)

ResAcc to Host Interaction Example



Rules

- Only the worst-case library is used for synthesis
  - 0p72vm40c
- The slack for setup and hold time should be non-negative
- Only the worst-case library is used in power analysis
  - 0p88v125c
- Optimization based on the specific test image is prohibited
- Model modification or re-train the weight is prohibited
- **The total latency of 2 images should be less than the total latency of 64 images / 16**

APR Specifications (1)

- Only the **macro layout** is needed
  - IO pads and bonding pads are not required
- **VDD** and **VSS power rings** should each be **2 µm wide**, with **5 ring pairs** required
- The IR drop of static power analysis should be less than or equal to **5%**

APR Specifications (2)

- Remember to add the **Power Rail** (follow pin)
- **Dummy metal layers** are not needed
- **Core Filler** must be added
- The **GDSII** file after APR must be generated
- Ensure that the APR DRC/LVS is completely **error-free**, and sign-off DRC/LVS check is not needed
- You can generate the IOC file first and then reload the file to set the pin position

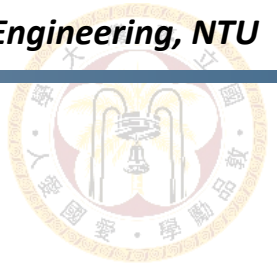
Simulation Settings

- Run the simulations using `01_run` in RTL/GateSim/PR
- You can only edit the bottom section of each `*_sim.f` file

```
1  ############################################################
2  # testbed, host, and dram
3  ############################################################
4  ../00_TESTBED/TESTBED.sv
5  ../10_AXI_DRAM/AXI_DRAM.sv
6  ../00_TESTBED/HOST.sv
7
8  ############################################################
9  # your design
10 ############################################################
11 resacc.v
12
13 ############################################################
14 # define
15 ############################################################
16 +define+TOTAL_IMAGE_NUMBER=64
17 +define+BATCH_IMAGE_NUMBER=16
18 +define+RESACC_CLK_PERIOD=1
19 +define+READ_LATENCY=12
20 +define+WRITE_LATENCY=16
21 +define+RTL
22 +define+DUMPFSDB
23
```

Final Project Schedule

- **Final Project Announcement:** April 8, 2025
- **Review Slide Submission: 12:00, April 29, 2025**
  - Each team should prepare a 10-minute presentation
  - Details are shown in page 31
- **Final Project Review:** April 29/May 6, 2025
- **Due: 14:00, June 3, 2025**
  - **No late submission is allowed**
- **Final Project Presentation:** June 10, 2025
  - Top-performing teams will be invited to present their design optimization strategies
  - Bonus points will be awarded for presentations



Review Slide Requirements

1. **The Data Format of Your Design**
   - Quantization method
   - Any data encoding method
2. **Hardware Implementation**
   - The block diagram of your design
   - The hardware design skills you are going to apply
3. **Other Design Skills**
   - Parallel processing
4. **Performance Estimated**
   - Area/Time/Power estimated based on your design
   - The improvement of your design

Review Slide Requirements

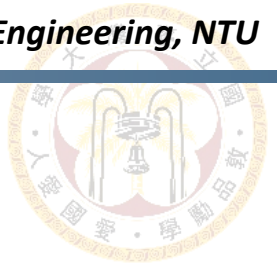
5. **The timeline for implementing your final project**

6. **Task Assignment for the Final Project of your team (if there is more than one member in your team)**

- **Submit the slide to NTU COOL before 12:00, April 29, 2025**
  - Both members in a team need to submit

Report Requirements

1. **APR Results**
   - Show the screenshot of the layout and DRC/LVS results
   - Show the screenshot of the IR drop analysis results
2. **Hardware Implementation**
   - Provide specifics about the hardware architecture, including the design of key modules and optimization techniques
3. **Performance Evaluation**
   - Total time for 64 images in RTL simulation
   - Layout area after APR
   - Power for 2 images in PR simulation



Design Spec File

- Please fill in your design spec

```
1   Team ID: xxxx
2   Student ID 1: xxxx
3   Student ID 2: xxxx
4   ----------------------
5   Clock period: 5.0 (ns)
6   Area: 50000.00 (um^2)
7   Total latency (64 images): 1000000.00 ns
8   Power (2 images): 10 mW
9   Is UPF used? (Yes/No)
10
```

- Clock period: the clock period in synthesis and simulation
- Area: the area after APR
- Total latency: the total latency in RTL simulation
- Power: the power in post APR simulation

Submission – Review Presentation

- Name the presentation slide to **teamID_final_v*k*.pdf or teamID_final_v*k*.pptx** (EX: team1_final_v1.pdf)

- Submit to NTU COOL
  - Both members of a team should hand it out to NTU COOL

- **Deadline:** 2025/4/29 12:00:00 (UTC+8)

Submission - Workstation

- Create a folder named **teamID_final (in lowercase)** (EX: team1_final) with the structure below:

```
team1_final/
├── 00_TESTBED/
│   └── model.dat
├── 01_RTL/
│   ├── resacc.v
│   ├── xxx.v (other verilog files, optional)
│   ├── xxx.sv (other systemverilog files, optional)
│   └── rtl_sim.f
├── 02_SYN/
│   ├── resacc_syn.tcl
│   ├── resacc_syn.sdc
│   ├── resacc_syn.ddc
│   ├── resacc_syn.area
│   ├── resacc_syn.timing_min
│   └── resacc_syn.timing_max
├── 03_GATE/
│   ├── resacc_syn.v
│   ├── resacc_syn.sdf
│   └── gate_sim.f
├── 04_APR/
│   ├── final
│   ├── final.dat
│   └── resacc.gds
├── 05_POST/
│   ├── resacc_pr.v
│   ├── resacc_pr.sdf
│   └── post_sim.f
├── 06_POST/
│   └── resacc_pr.power
├── 07_UPF/ (optional)
│   ├── resacc.rtl.upf
│   ├── resacc.syn.upf
│   └── resacc.apr.upf
└── design.spec
```

Submission - Workstation

- Pack the folder **teamID_final** (EX: team1_final) into a tar.gz file named **teamID_final.tar.gz** (EX: team1_final.tar.gz)
  - `tar -zcvf teamID_final.tar.gz teamID_final`
  - All letters are in lowercase.
  - Put the tar.gz file at the root of both your ADFP accounts

- TA will only check the latest version

- **Deadline:** 2025/6/3 14:00:00 (UTC+8)

Submission – NTU Cool

- Create a folder named **teamID_final (in lowercase)** (EX: team1_final) with the structure below:

```
team1_final/
├── 01_RTL/
│   ├── resacc.v
│   ├── xxx.v (other verilog files, optional)
│   └── xxx.sv (other systemverilog files, optional)
├── 98_ALGORITHM/
│   ├── dat_gen.py/dat_gen.ipynb
│   └── model.dat
├── report.pdf
└── design.spec
```

- Compress the folder **teamID_final (all lowercase)** in a tar file named **teamID_final_v*k*.tar.gz** (EX: team1_final_v1.tar.gz)
  - **`tar -zcvf teamID_final_v*k*.tar.gz teamID_final`**
  - Both members of a team should hand it out to NTU COOL
- **Deadline:** 2025/6/3 14:00:00 (UTC+8)
- Ensure all required files are submitted

Performance Index

1. **Performance index: Area x Time x Power x (Error Rate - 0.1)**
2. **Area:** Layout area (die area in um^2)
3. **Time:** Total simulation time for the **64 images** (in ns)
4. **Power:** Simulation power for **2 images** after APR (in mW)
5. **Error Rate - 0.1:** The error rate of the 64 test image - 0.1
   - Example: accuracy = 0.7, error rate = 0.3, error rate - 0.1 = 0.2
6. **Baseline: Time < $5x10^6$, Area < $5x10^5$, Error Rate < 0.37**
7. **Goal: Time < 10^6, Performance index < $5x10^{11}$, Pass the baseline**

Grading Policy

- Implementation 40% + Goal 20% + Performance 30% + Review Presentation & Report 10%

Item	%	Description
RTL Simulation	20	Pass 64 pattern simulations
Synthesis	10	Pass gate-level simulation
APR	10	Finish APR with no DRC/LVS errors Pass post-layout simulation
Goal	20	Pass the goal requirement
Performance	30	Linear score between all teams passing the goal
Presentation	5	Details are shown in page 31
Report	5	Details are shown in page 33

Grading Policy

- **No late submission is allowed**
  - Any submission after the deadline will receive 0 points
- **5-point deduction** for incorrect naming or format
  - Pack all files into a single folder and compress the folder
  - Ensure that the files submitted can be decompressed and executed without issues
- **No plagiarism**
  - Plagiarism in any form, including copying from online sources, is strictly prohibited

Discussion

- **NTU Cool Discussion Forum**
  - For any questions not related to assignment answers or privacy concerns, please use the NTU Cool discussion forum
  - TAs will prioritize answering questions on the NTU Cool discussion forum

- **Email:** f12943014@ntu.edu.tw
  - Title should start with [113-2 ACVSD Final Project]
  - Email with wrong title will be moved to trash automatically

Reference

- [1] ResNet: Residual neural network – Wikipedia
- [2] Cifar-10: CIFAR-10 and CIFAR-100 datasets
- [3] CNN and Convolution Layer
  - ACVSD Midterm
  - 【機器學習2021】卷積神經網路 (CNN)
- [4] AXI Protocol
  - ACVSD Lecture
  - AMBA AXI Protocol Specification